Lesson Planning

Complete Lesson Plan Book

Class: FSc Pre-Engineering · Class 12

Subject: Mathematics

Medium: English

Academic Year 2026-27

This book is prepared as a teaching guide for teachers.

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Learning Objectives

• Students will define a matrix and identify its order, rows and columns.

• Students will classify matrices as row, column, square, diagonal, scalar and identity matrices.

• Students will find the determinant of a 2x2 matrix and explain what it tells us about the matrix.

Key Concepts

• A matrix is a rectangular arrangement of numbers in rows and columns, written inside square brackets.

• The order of a matrix is written as m x n, where m is the number of rows and n is the number of columns.

• A square matrix has the same number of rows and columns; an identity matrix has 1s on the main diagonal and 0s elsewhere.

• The determinant of a 2x2 matrix [[a, b], [c, d]] is ad - bc.

• A matrix with determinant zero is called singular; a matrix with non-zero determinant is non-singular.

Teaching Methodology (Step by Step)

1. The teacher writes a 2x3 table of marks of three students in two subjects on the board and asks how such a table can be written compactly.

2. The teacher introduces the word matrix and writes the marks table as a matrix, pointing out rows, columns and entries.

3. The teacher explains the order of a matrix with two or three examples and asks students to state the order of given matrices.

4. The teacher presents row, column, square, diagonal, scalar and identity matrices one by one with a small example of each.

5. The teacher derives the determinant of a 2x2 matrix as ad - bc and works out two numerical examples on the board.

6. The teacher explains singular and non-singular matrices and asks students to test given 2x2 matrices.

Classroom Activities

• Order race: the teacher shows five matrices and students write down the order of each; the fastest correct row wins.

• Matrix hunt: students find one example each of a square matrix and a diagonal matrix from the textbook exercises.

• Determinant drill: in pairs, students compute the determinants of six given 2x2 matrices and check each other's answers.

Assessment Questions

1. Define a matrix and write the order of [[2, 5, 1], [0, 3, 4]].

2. Give one example each of a row matrix, a column matrix and an identity matrix of order 3.

3. Find the determinant of [[3, 2], [1, 4]] and state whether the matrix is singular or non-singular.

Homework

• Write five matrices of different orders and state the order of each.
• Find the determinants of the 2x2 matrices given in textbook exercise 3.1, questions 1 to 5.

Time Allocation

40 minutes

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